

IP Addressing Plan

Ubuntu Innovations (Pty) Ltd
Week 2 – Task 5 | IT Support Capstone Project

1. Network Overview

Ubuntu Innovations (Pty) Ltd operates a segmented office network using the private IP range 192.168.0.0/16. The network is divided into five VLANs, each assigned to a specific device category. This segmentation improves security, simplifies traffic management, and allows department-level access control.

The router/firewall serves as the central gateway at 192.168.1.1 and manages all inter-VLAN routing and internet access.

2. Subnet Allocation Table

The following subnets have been assigned to support all 25 employees and network devices across five departments:

Device Type	Subnet	IP Range	Gateway	Usable Hosts
Servers / NAS	192.168.10.0/24	192.168.10.1 – 192.168.10.254	192.168.10.1	253
User Devices	192.168.20.0/24	192.168.20.1 – 192.168.20.254	192.168.20.1	253
Printers	192.168.30.0/24	192.168.30.1 – 192.168.30.254	192.168.30.1	253
Wi-Fi (Wireless)	192.168.40.0/24	192.168.40.1 – 192.168.40.254	192.168.40.1	253
Management / Admin	192.168.50.0/24	192.168.50.1 – 192.168.50.254	192.168.50.1	253

3. Static IP Assignments

The following infrastructure devices have been assigned static IP addresses to ensure consistent network identity and reliable connectivity:

Device	IP Address	Notes
Router / Firewall	192.168.1.1	Main gateway for all traffic
Managed Switch	192.168.1.2	Admin management interface
WAP 1	192.168.1.3	Wireless Access Point – Zone A
WAP 2	192.168.1.4	Wireless Access Point – Zone B
NAS Device	192.168.10.10	File storage and backups
Printer 1	192.168.30.10	Shared laser printer
Printer 2	192.168.30.11	Shared laser printer

4. DHCP Configuration

User desktops, laptops, and wireless devices receive IP addresses dynamically via DHCP. A portion of each subnet is reserved for future static assignments or infrastructure expansion.

Subnet	DHCP Range	Excluded (Static Reserved)
User Devices	192.168.20.50 – 192.168.20.200	.1 – .49 reserved for static devices
Wi-Fi	192.168.40.50 – 192.168.40.200	.1 – .49 reserved for static devices

5. DNS Configuration

Internal DNS resolution is handled by the NAS/server at 192.168.10.10. External DNS queries are forwarded to the following public resolvers for redundancy:

- Primary: 8.8.8.8 (Google Public DNS)
- Secondary: 1.1.1.1 (Cloudflare DNS)

6. Design Justification

The use of separate subnets for each device category aligns with IT security best practices. Key benefits include:

- Isolation: Printer and server traffic is separated from general user traffic, reducing exposure to lateral movement attacks.
- Scalability: Each /24 subnet supports up to 253 hosts, providing ample room for future growth.
- Manageability: Static IPs on infrastructure devices ensure reliable remote management and monitoring.
- Policy enforcement: VLAN segmentation enables firewall rules to restrict inter-subnet access where required.